



THE AI TOOLBOX · SPECIAL PROJECT CLASS

Your First Research Poster

IEEE HealthCom 2026 · Young Researcher Symposium

One student, one poster. We walk the **whole AI-era research pipeline** — from question to a real conference submission — with every tool you've already learned in Classes 1–10.

Merit Point Academy · 科研海报冲刺 · press **S** notes · → next ·  to comment



A real IEEE conference is calling for **your** poster

The graphic features a background image of the New York City skyline with the Empire State Building and the Brooklyn Bridge. At the top left, it says 'IEEE HealthCom 2026' with a Wi-Fi icon. To the right, 'New York City, USA' is written in a handwritten style. The main title 'YOUNG RESEARCHER SYMPOSIUM (YRS)' is in large, bold, blue letters. Below it, a dark blue button says 'POSTER SESSION' with a calendar icon. Text below the button reads: 'Showcase your research. Get feedback. Build connections. Shape the future of eHealth.' To the right of this text is a circular badge with a calendar icon and the text 'IEEE HealthCom 2026 October 19–21, 2026'. Below the main text are four icons in circles: a person at a presentation board, a group of people, two hands shaking, and a trophy. Each icon has a label: 'PRESENT your research', 'GET FEEDBACK from experts', 'CONNECT with researchers', and 'AWARDS for outstanding posters'. At the bottom, there are three boxes: 'POSTER PROPOSAL SUBMISSION DEADLINE AUGUST 15, 2026' with a calendar icon, 'ACCEPTANCE NOTIFICATION AUGUST 31, 2026' with an envelope icon, and 'SYMPOSIUM DATE During IEEE HealthCom 2026 OCTOBER 19–21, 2026' with a group of people icon. At the very bottom, there is a dark blue button that says 'SUBMIT YOUR POSTER PROPOSAL!' with a pencil icon, and below it, the URL 'https://edas.info/N34385'. To the right of the button is a QR code with an arrow pointing to it and the text 'Scan to submit!'. At the bottom left, there is a small globe icon and the text 'Learn more: https://healthcom2026.ieee-healthcom.org/authors/call-posters-young-researcher-symposium'.

IEEE HealthCom 2026 · Young Researcher Symposium (YRS)

A **poster session** at a real IEEE conference on eHealth — held in **New York City, October 19–21, 2026**. Young researchers present a poster, get feedback from experts, connect with researchers, and compete for **outstanding-poster awards**.

The three official dates 官方时间线

Poster proposal deadline · **Aug 15, 2026**

Acceptance notification · **Aug 31, 2026**

Symposium · **Oct 19–21, 2026**

Submit at **edas.info/N34385**: an **extended abstract**, up to **2 pages** — including **figures, tables AND references** — in the **IEEE conference template** · first author must be a high-school/undergrad **student** · at least one **faculty mentor as co-author** · accepted posters are expected to be **presented in person** · details: healthcom2026.ieee-healthcom.org → Authors → Call for Posters (YRS)



READ THIS FIRST · 最重要的一页

The goal is the **process**, not perfection

This project's purpose is to walk one full lap of the **AI-era research workflow** — question → literature → data → analysis → poster → presentation — so that "doing research" stops being a mystery and becomes something you have **done**.

Knowledge is endless 知识学不完

Nobody — not your mentor, not any professor — has "finished" learning. Missing background is **normal**, not a reason to wait.

The one real requirement

Everything **on your draft**, you can **explain clearly** in your own words. That's the whole bar. 只要能把自己 draft 里涉及的部分讲清楚即可。

Don't panic, ask 不着急

Stuck on a concept your poster needs? Shrink the claim, or bring it to your mentor / the class. Both are researcher moves.

For students AND parents 给学生和家长 : this project is how a student **enters the research world** — by doing one honest, complete, small project. It is not a test of how much you already know. 这次主要是带孩子走一遍 AI 时代的科研流程，帮助孩子进入科研领域；具体知识没有完全掌握不必着急。



WHAT A POSTER IS · 海报的解剖学

One big page that tells one complete story

A research poster is your whole project on a single large page (typically **A0 / 36×48 in, portrait**), designed so a stranger gets the story in **90 seconds** — then talks to you about it.

1 · Header

Title (your question, in plain words), your name, school, program. Big — readable from 3 m away.

2 · Intro & Question

Why this topic matters + your **one research question** (the input → output → metric recipe from Class 2).

3 · Data

Where the data came from, what one row means, size, license — your Class 7 **datasheet**, compressed.

4 · Methods

What you did, as a simple pipeline diagram: clean → split → model/analyze (Classes 8–10). Pictures beat paragraphs.

5 · Results

1–3 honest charts (Class 9 rules: labeled axes, no lying axes) + the honest numbers (test score, not train score).

6 · Conclusion & Limits

What you found, what it does **not** prove, what you'd do next. A limitations line makes you **MORE** credible, not less.

Rule of thumb: ≤ 800 words total on the whole poster · every section skimmable · at least one figure per column · references + QR code in a small footer.



OUR REQUIREMENTS · 本项目要求

One student, one poster — the checklist

Non-negotiables 硬性要求

- ✓ **One poster per student** — your own question, your own data, your own name on it
- ✓ All six sections present (title → conclusion)
- ✓ A **real dataset** you actually analyzed (Classes 7–10 work counts!)
- ✓ At least **one chart you made** from real data, axes labeled
- ✓ Honest numbers: test-set results, limitations stated
- ✓ You can **explain every element** on the poster in your own words

Format 规格

- 📏 A0 portrait (or 36×48 in) — one page, PDF
- 📄 Title ≥ 72 pt · headings ≥ 40 pt · body ≥ 24 pt
- 📊 Figures first: aim for ~40% figures, ≤ 800 words
- 🎨 2–3 colors max, dark text on light background
- ✂️ Build in Google Slides / PowerPoint / Canva / LaTeX — set the page size **FIRST**
- 📎 Footer: references + a QR code to your project page (Guide 04!)

Topic note: submissions must align with the **HealthCom scope — health-related**. The official topics are broad: AI/ML & data analytics for healthcare · wearables & health monitoring · telemedicine & digital-health apps · biomedical signals, imaging & health data · eHealth security/privacy · student-led health-tech prototypes. Almost any interest has a health angle (sleep, screen time, sports data, air quality, mental health...) — your mentor helps you find it in proposal week, *or* aims the same poster at a non-health venue instead.



YOUR TOOLBOX · 你已经有全部工具

Every poster section has an AI co-pilot you've already met

Poster step	AI / tool	Where you learned it
Sharpen the question	ChatGPT/Claude as question critic (input → output → metric)	Class 2 · Class 4
Related work, 3–5 papers	Elicit · Semantic Scholar · Connected Papers	Class 6
Find & judge a dataset	Kaggle / UCI + the RIGHT checklist + datasheet	Class 7 · Guide 05
Clean + analyze + model	pandas · matplotlib · scikit-learn in Colab, AI assistant beside you	Classes 5, 8, 9, 10
Draft the text (then verify!)	LLM drafts abstract/intro from YOUR results; you fact-check every claim	Class 4
Figures & layout at pro quality	Claude Code: regenerate charts, tune styling, even lay out in LaTeX	Class 10 demo · Guide 06

Using AI is not optional here — it's the design. 使用 AI 不是可选项，而是课程设计本身。 This project and its assignments are built for the AI-era research workflow: they are scoped assuming an AI assistant at every step, and would be **much harder without one**. Learning to collaborate with AI — prompt, verify, own — is itself a core skill this course trains. 学生完成作业必须使用 AI 工具，这正是顺应时代趋势、培养与 AI 合作的能力。



USING AI HONESTLY · 诚实地用 AI

AI may draft. **You must own.**



Great uses

Brainstorm titles · critique your question · summarize a paper you then skim · explain YOUR error message · draft a section from YOUR real results · polish grammar · suggest chart improvements.



Never

Invented data or results · citations you never opened (hallucination risk — Class 4!) · numbers on the poster that don't come from your own notebook · text you can't explain when asked.

The ownership test

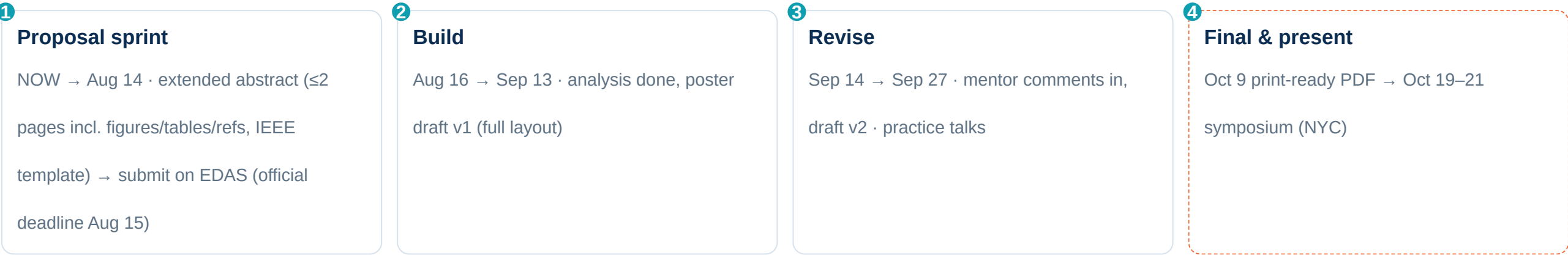
Before anything goes on the poster, close the chat window and explain it out loud. Can't? Then it's not yours yet — learn it or cut it.

The 2-minute verify routine (Class 4) applies to every AI sentence: check the claim against your own notebook output or the actual paper. At the symposium, experts will ask follow-up questions — the poster is a promise that **you** stand behind every pixel.



THE ROADMAP · 总时间线

Ten weeks, four milestones



Official dates 官方节点

Proposal due **Aug 15** · acceptance **Aug 31** · symposium **Oct 19–21**. Ours are set a day+ earlier — never surf the deadline.

Weekly heartbeat

Every class starts with your 3-minute progress presentation (next slide). Small steps every week beat a heroic final weekend.

If not accepted?

The poster still gets finished, presented in class, and aimed at the next venue — science fairs and other symposia accept the same poster. Zero wasted work.



EVERY CLASS FROM NOW ON · 课堂新惯例

The progress-presentation ritual

From now until the symposium, **every class opens with every student presenting ~3 minutes** on their poster's current state. This is not a status check — it's **presentation training**, the exact skill the poster session tests.

Your 3 minutes · 三句话结构

- 1 **What changed** since last class — show the thing (chart, section, draft screen)
- 2 **Where I'm stuck** — one concrete blocker, asked as a question
- 3 **Next step** — the one thing you'll do before next class

Mentor review timeline 导师评审节奏

-  Live comments on every progress talk, in class
-  **Written review comments within 48 h** of each milestone (proposal, v1, v2)
-  Final check pass on your print-ready PDF before it goes to print

Presenting "I'm stuck" is a WIN. Naming a blocker precisely is a research skill; hiding it for two weeks is the only real failure mode. 卡住了大方说出来 —— 这正是科研训练的一部分。



What each milestone must contain

Milestone	Due (11:59 pm ET)	Must contain	Mentor comments by
Proposal draft	Wed Aug 12	Extended-abstract draft (IEEE template, ≤ 2 pages <i>incl. figures, tables & references</i>): title · question · dataset · planned analysis	Aug 13 (fast lane!)
EDAS submission	Fri Aug 14	Submitted at edas.info/N34385, mentor listed as co-author · confirmation screenshot	— (official deadline Aug 15)
Poster draft v1	Sun Sep 13	Full A0 layout, all 6 sections drafted, ≥ 1 real chart from your data	Sep 16
Poster draft v2	Sun Sep 27	Every v1 comment addressed · final figures · practice-talk ready	Sep 30
Final print PDF	Fri Oct 9	Print-ready A0 PDF · references + QR footer · ownership test passed	Final check Oct 10–12, then print

These five milestones are posted as **assignments with due dates** at academic.hopeembark.org/assignments — submit there, and the usual reminder emails apply. 以上里程碑已作为作业发布，附截止时间与邮件提醒。



QUICK CHECK · THE SPIRIT OF THIS PROJECT

Quick check · tap an answer

Your draft uses linear regression, and honestly you're shaky on how gradient descent works inside. The best move is:

- A** Delete the model from the poster — better safe than sorry
- B** Keep it, and hope nobody at the poster session asks
- C** Panic-study all of machine learning before touching the poster again
- D** Be able to say clearly what regression DOES for your question (fits a line to predict Y from X , scored on test data) — and revisit the Class 10 deck or ask your mentor for the inside details



SUPPORT · 需要的支持

What you need — and where it comes from

From yourself 学生

~3–4 focused hours/week · show up to the ritual every class · submit milestones on time · own every word on your poster.

From your mentor 导师

Topic & dataset triage · written comments within the review windows · practice-talk coaching · final print check. Ask early, ask specifically.

From your family 家长

Protected weekly work time · printing the final A0 poster (~\$40–80 at a local print shop) · and the New York question: **if accepted, the symposium expects at least one student author to present in person** — so talk about the October trip as a family **before** we submit.

家长请放心：本项目的核心价值在于完整走一遍科研流程——提出问题、找数据、分析、做海报、公开讲述。孩子不需要提前掌握全部知识，只需把自己海报上涉及的内容讲清楚。课程与作业均按 **AI 时代的研究方式** 设计：孩子需要（也应当）使用 AI 工具来完成——这不是走捷径，而是刻意培养“与 AI 协作”这项新时代的核心能力。另请注意：若海报被录取，会议原则上要求至少一名学生作者现场展示，请在提交前（8月14日前）全家商量好十月纽约行程；无论最终结果如何，这份海报和这段训练都完整属于孩子。



The proposal sprint starts **now**

- Pick your question (steal it from your Class 2/7 homework — that's what it was for)
- Name your dataset and check its license (Guide 05 checklist)
- Draft your extended abstract — IEEE template, ≤ 2 pages **including figures, tables & references**; start from a 150-word core, then grow it. AI may help draft, you must own every sentence
- Submit the proposal draft by **Wed Aug 12** → mentor comments within 24 h
- Submit on EDAS (**edas.info/N34385**) by **Fri Aug 14**
- Come to next class ready for your first 3-minute progress talk